

Grade 7
Unit 6
Lesson 7 Practice

Name _____
Date _____
Period _____

Solve each problem using proportional reasoning. Show your work.

1. Sashia, a competitive swimmer, swims at a constant speed of 50 meters 54 seconds. To the nearest tenth of a second, how long does it take her to swim 235 meters?

$$\frac{50 \text{ m}}{54 \text{ sec}} = \frac{235 \text{ m}}{x}$$

253.8 seconds

2. To make a shade of paint called lime green, mix 7 quarts of yellow paint with 4 cups of blue paint. How much yellow paint should be mixed with 6.5 cups of blue paint to make lime green?

$$\frac{7 \text{ yellow}}{4 \text{ blue}} = \frac{x}{6.5}$$

11.375 quarts yellow

3. It takes an ant farm 2 days to consume $\frac{1}{4}$ of an apple. At that rate, in how many days will the ant farm consume 5 apples?

$$\frac{2 \text{ days}}{0.25 \text{ apple}} = \frac{x}{5}$$

40 days

4. Raspberries cost \$2.98 per $\frac{1}{2}$ pint. How much would it cost to purchase 3 pints of blackberries?

$$\frac{\$2.98}{0.5 \text{ pint}} = \frac{x}{3 \text{ pints}}$$

\$17.88

5. Thiago is going on a road trip with his family. He wants to find out how long he will be in the car. His mother told him that their destination is 1,980 miles away and she will be driving at a constant 65 miles per hour. To the nearest tenth of an hour, how long will it take Thiago and his family to reach their destination?

$$\frac{65 \text{ miles}}{1 \text{ hr}} = \frac{1980 \text{ mi}}{x}$$

30.5 hrs

6. An 14-inch pizza from Pizza Pi is cut into 8 slices and costs \$11.95. Rita has enough pizza for each person to eat 3 slices. If Rita has \$107.55 worth of pizza, how many people does she feed?

$$\frac{8}{\$11.95} = \frac{x}{107.55}$$

$$72 \div 3 = 24 \text{ people}$$

72 slices

7. Almonds cost \$47.45 per 5 pound bag. How much would a 3 pound bag of cashews cost?

$$\frac{\$47.45}{5 \text{ lb}} = \frac{x}{3 \text{ lb}}$$

\$28.47

8. It takes Richard 45 minutes to mow $\frac{1}{3}$ of an acre of grass. If his yard is 1.2 acres, how many hours will it take him to mow the lawn?

$$\frac{45}{60} = 0.75 \text{ hour}$$

$$\frac{0.75 \text{ hour}}{x} = \frac{\frac{1}{3} \text{ acre}}{1.2 \text{ acres}}$$

2.7 hours

9. Randi wants to paint the rooms in her house. It takes her 30 hours to paint all 5 rooms with 1 coat of paint each. How many rooms can she paint each day if she can spend 6 hours a day painting?

$$\frac{30 \text{ hrs}}{5 \text{ rooms}} = \frac{6 \text{ hrs}}{x}$$

1 room

10. Typing at a rate of 45 words per minute is considered a good speed. If a student has to type a paper that contains 900 words, how many minutes will it take the student to type the paper?

$$\frac{900}{45} = 20 \text{ minutes}$$